

Jason Yi

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EDUCATION

University of North Carolina at Chapel Hill

Chapel Hill, NC

Bachelor of Science in Computer Science, Statistics | Dean's List

Exp Grad: May 2027

- **TA:** Algorithms (F25, S25), System Fundamentals in C (F24), Data Structures and Analysis in Java (F23, S24)
- **Research:** Optimized TreeFARMS algorithm in **C++/ Python** to produce **Decision Trees** under Chudi Zhong
- **Coursework:** Operating Systems, Machine Learning, Algorithms, Databases, Stochastic Modeling, Probability

EXPERIENCE

Google

May 2026 - August 2026

Software Engineer Intern

New York, NY

- Incoming Software Engineering Intern - Team: **Fuchsia OS** in **Rust** and **C++**

Scale AI

October 2025 – May 2026

Technical Advisor Intern - GenAI

Remote

- Built a **C++** validation pipeline to enforce schema compliance and data integrity across open-source training corpora, directly supporting quality assurance for **LLM** reasoning and code generation benchmarks
- Evaluated frontier **LLM** performance on the **MCP** by designing adversarial, multi-step prompt trajectories targeting failures in **JSON** function calling, tool selection, and external API integration across model families

Amazon Web Services

May 2025 – August 2025

Software Development Engineer Intern

Seattle, WA

- Built a scalable mock data generation system for **AWS Compute Optimizer** in **Kotlin** and **Python** with **AWS AppConfig**, **AWS Lambda**, and **S3** to reduce false **Sev-2** alerts by decoupling canary tests from upstream data
- Accelerated bug bash validation by **20%** by integrating a **GenAI**-driven agentic workflow to automate ingestion of schema-validated mock data into existing indexing pipelines to streamline testing and reduce operation delays

Fidelity Investments

June 2024 – August 2024

Software Engineer Intern

Durham, NC

- Developed Backend services in **GraphQL** via **Experience API** for **Account Opening** which impacts **50+ million users**, and Frontend services in **Angular** and **TypeScript** for Crypto IRA
- Implemented customer info, address validation, and risk analysis services to prevent user fraud or illegal activity during account opening using **TypeScript** and **GraphQL** by matching data from multiple downstream APIs

PROJECTS

AutoReturns 🌀 | *Python, PyTorch, Scikit-learn, NumPy, Pandas, Seaborn, Matplotlib*

- Analyzed **30M+** rows of U.S. stock return data (**CRSP**, **S&P 500**, **Russell 3000**) to document stylized facts (e.g., volatility clustering, fat tails) using **PCA**, **autocorrelation**, and **return histograms**
- Reduced dimensionality of stock return data by **90%** using kernel PCA in **scikit-learn** and autoencoders in **PyTorch**, preserving **≥85%** variance and uncovering latent relationships across asset classes
- Clustered PCA and autoencoder embeddings with **KMeans** to identify sectoral and macroeconomic groupings

Synapse: Distributed AI Scheduler 🌀 | *Go, gRPC, Protocol Buffers, Linux Kernel*

- Engineered a fault-tolerant **Go** scheduler implementing **Gang Scheduling** to enforce atomic resource allocation across dynamic worker nodes, eliminating deadlocks common in **distributed multi-GPU training** workloads
- Developed a bi-directional **gRPC** control plane to orchestrate a custom **Rust** container runtime (**Carapace**), bridging high-level scheduling with low-level **Linux syscall** execution via the **Fork/Exec** model

Carapace: Linux Container Runtime 🌀 | *Rust, C++, Linux Syscalls, Cgroups, Namespaces, Bash*

- Architected a container runtime in **Rust** that interfaces directly with the **Linux Kernel** via **syscalls**, leveraging **Namespaces** (UTS, PID, Mount) and **chroot jails** to achieve strict process isolation
- Enforced resource quotas via **Control Groups (Cgroups v2)** to prevent process exhaustion attacks (e.g., fork bombs), and developed a **C++ FFI** interoperability layer to bridge Rust with low-level system inspection tools

TECHNICAL SKILLS

Languages: C/C++, Go, Python, Rust, Java, Kotlin, TypeScript, JavaScript, HTML/CSS, SQL, Swift/SwiftUI

Frameworks/Libraries: PyTorch, Tensorflow, NumPy, Pandas, Matplotlib, React.js, GraphQL, Node.js

Developer Tools: AWS, Kubernetes, Git, GitHub, Linux Kernel, Vim, Jira, Jenkins, XCode